Group Presentation

Use Operations Research to Solve Problem
A case study of ICM 2024

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Introduction: Water Level Adjust

Problem

- The Great Lakes of the United States and Canada are the largest group of freshwater lakes in the world. The five lakes and connecting waterways constitute a massive drainage basin that contains many large urban areas in these two countries, with varied climate and localized weather conditions.
- These levels are the result of complex interactions among temperature, wind, tides, precipitation, evaporation, river flows and runoff, reservoir policies, seasonal cycles, and long-term climate changes.
- While the two control dams, many channels and canals, and the drainage basin reservoirs may be controlled by humans, the rates of rain, evaporation, erosion, ice jams, and other water-flow phenomena are beyond human manipulation.
- There requests support from your company to assist with management and models for the control mechanisms that directly influence water levels in the Great Lakes flow network.

Assumption

- Water Flow between Lake Michigan and Lake Huron does not influence our model
- Consider the average water level of historical data as the initial value for the optimal water level, then iterate to seek an improved solution
- The inflow between each lake is equal to the outflow from the next lake
- Do not consider other runoff outside the Great Lakes
- Stakeholders only consider Lake Ontario (because the interests of Lake Ontario have the greatest impact)

Operations Research

Question 1: Find the Optimum Water Level

Objective Function

The goal is to minimize the weighted absolute deviation of lake levels from their ideal levels:

$$\min \sum_{i=1}^{n} w_i \cdot |x_i - L_i| \tag{1}$$

where

- x_i: current water level of lake i (decision variable)
- L_i: ideal water level of like i
- w_i: weight assigned to priority of stakeholder constraints (decending order: $5, 4, 3, 2, 1$)
- n: total number of lakes

Constraints

Residents' Constraint

The water level should be maintained at ± 0.5 meters of the mean water level to avoid flooding or dry seasons

$$L_i - 0.5 \leq x_i \leq L_i + 0.5, \quad \forall i \in \{1, 2, \ldots, n\}$$

Commercial Shipping Companies' Constraint

The water level must be at least 1.5 meters to ensure safe passage of ships, but not more than 2.5 meters to avoid damage to shore facilities.

$$1.5 \leq x_i \leq 2.5, \quad \forall i \in \{1, 2, \ldots, n\}$$

Constraints *(Continued ii.)*

Power Generation Companies' Constraint

The water level needs to be maintained within the range that allows efficient power generation, for example above 1.8 meters but not more than 2.2 meters.

$$1.8 \leq x_i \leq 2.2, \quad \forall i \in \{1, 2, \ldots, n\}$$

Recreational Activities' Constraint

The water level should be maintained within ± 0.3m of the mean water level to facilitate recreational activities.

$$L_i - 0.3 \leq x_i \leq L_i + 0.3, \quad \forall i \in \{1, 2, \ldots, n\}$$

Constraints *(Continued iii.)*

Environmental Protection Organizations' Constraint

The water level should be maintained at no less than 1.4 meters to support the health of the lake ecosystem.

$$1.4 \leq x_i, \quad \forall i \in \{1, 2, \ldots, n\}$$

- These values are set to balance the interests of all parties in the management of water levels, while taking into account operability and environmental protection.
- In actual application, adjustments need to be made based on hydrological studies, environmental assessments and socio-economic analyses to ensure a balance of interests.

Linear Modeling

Objective:

$$\min \sum_{i=1}^{n} w_i \cdot |x_i - L_i|$$

Subject to:

$$\begin{aligned} L_i - 0.5 \leq x_i \leq L_i + 0.5 \\ 1.5 \leq x_i \leq 2.5 \\ 1.8 \leq x_i \leq 2.2 \\ L_i - 0.3 \leq x_i \leq L_i + 0.3 \\ 1.4 \leq x_i \end{aligned}$$

Operations Research

Question 2: Algorithm to Maintain Optimal Water Levels

Network Model

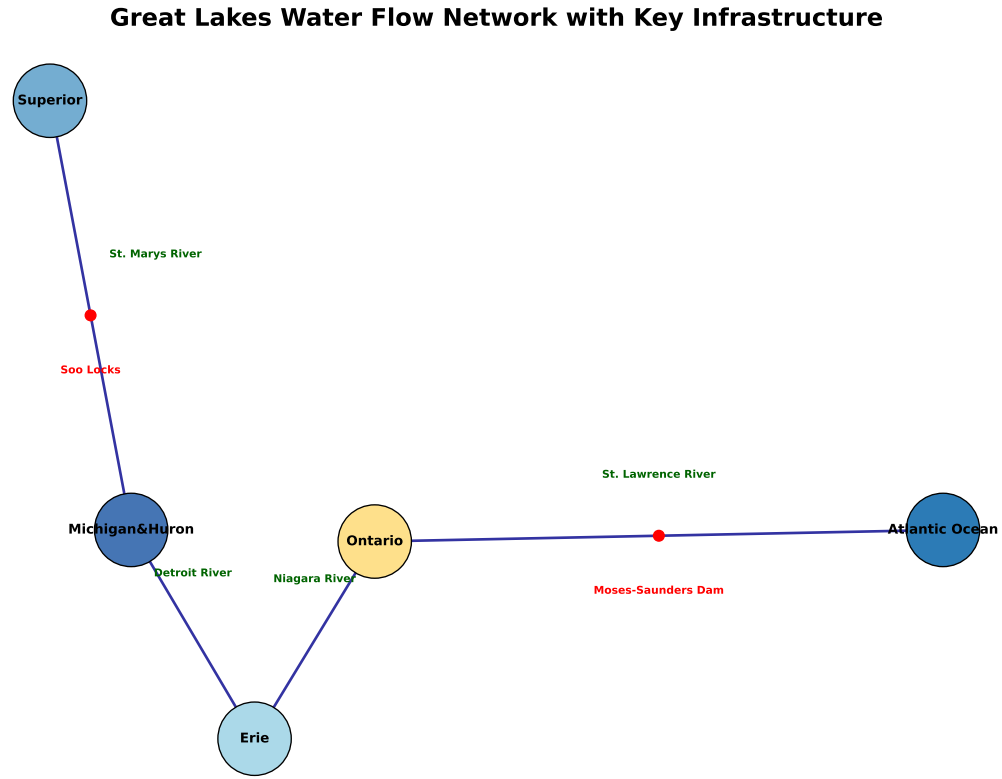


Figure: Network Diagram

Historical Average Water Level

Let D be the dataset containing historical water level data for each lake over different years. Let S be the set of sources (lakes) in the dataset. For each source $s \in S$, the historical average water level $\bar{w}_s$ is calculated as:

$$\bar{w}_s = \frac{1}{N_s} \sum_{i=1}^{N_s} w_{s,i} \tag{2}$$

where

- N_s: number of years of data available for lake s
- $w_{s,i}$: water level of lake s in year i

Ideal Water Level

Assume the ideal water level for each lake is the overall average of the historical average water levels across all lakes. Let $\bar{w}$ be the ideal water level, calculated as:

$$\bar{w} = \frac{1}{|S|} \sum_{s \in S} \bar{w}_s \tag{3}$$

where

- S: the set of all the lakes
- $|S|$: the number of whole lakes

Sum of Differences from Ideal Water Level

For each lake $s \in S$, the difference from the ideal water level Δw_s is:

$$\Delta w_s = \bar{w}_s - \bar{w} \tag{4}$$

The sum of these differences $\Sigma \Delta w$ is:

$$\Sigma \Delta w = \sum_{s \in S} \Delta w_s = \sum_{s \in S} (\bar{w}_s - \bar{w}) \tag{5}$$

- These formulas capture the process of calculating the historical averages, determining the ideal water level, and computing the total deviation from this ideal level.

$$\text{Objective} = \sum_{i=1}^{n} \left| \text{ideal_levels}[node_i] - \frac{\text{flows}[i]}{\text{max_flow_capacity}[river_i]} \right| \tag{6}$$

Subject to

$$0 \leq \text{flows}[i] \leq \text{max_flow_capacity}[river_i] \quad \forall i \tag{7}$$

where:

- n: number of nodes (or rivers).
- $node_i$: the i-th node.
- river_i:is the i-th river.
- $\text{flows}[i]$: flow of the i-th river.
- $\text{ideal_levels}[\text{node}_i]$ is the ideal level of the i-th node.
- $\text{max_flow_capacity}[\text{river}_i]$ is the maximum flow capacity of the i-th river.

Result

Lake	Water Level
Lake Erie - Mean Water Level	174.280761
Lake Michigan and Lake Huron - Mean Water Level	176.328913
Lake Ontario - Mean Water Level	74.829783
Lake St. Clair - Mean Water Level	175.103043
Lake Superior - Mean Water Level	183.346413

Table: Ideal Lake Level

Month	Marys River Flow	Lawrence River Flow
Jan.	1994.678935	7256.834807
Feb.	1944.740158	7251.686290
Mar.	1939.986974	7890.102407
	⋮	
Dec.	2041.179238	7631.389424

Table: River Flow

Operations Research

Question 3: Sensitivity Analysis in Environment Conditions

Objective Function

The goal is to minimize the total deviation of river flows from ideal lake levels

$$\min_{\mathbf{F}} \sum_{i=1}^{N} \left| L_i - \frac{F_i}{C_i} \right| \tag{8}$$

where

- N: number of rivers (e.g., 2: St. Marys River and St. Lawrence River)
- L_i: ideal level for the lake associated with river i (e.g., Superior or MHEO)
- F_i: flow rate of river i
- C_i: maximum flow capacity of river i

Constraints

Flow Bounds

The flow rates must be physically feasible

$$0 \leq F_i \leq C_i \quad \forall i \in 1, 2$$

No Environment Influence

$$F_i = \text{Historicalflowrates} \tag{9}$$

Increased Precipitation

$$F_i = F_i^{\text{historical}} \times (1 + p_i) \tag{10}$$

Increased Evaporation

$$F_i = F_i^{\text{historical}} \times (1 + e_i) \tag{11}$$

Adjusted Objective Under Environmental Impacts

For Adjusted flows F_i^{adjusted}, the objective becomes

$$\min_{\mathbf{F}} \sum_{i=1}^{N} \left| L_i - \frac{F_i^{\text{adjusted}}}{C_i} \right|$$

where

- F_i^{adjusted}: computed based on the environmental condition (e.g., increased precipitation rate p or evaporation e)

Modeling

Water Level Change (Velocity function)

$$v_t = \frac{\mathrm{d}h_i}{\mathrm{d}t} = I(t) - O(t) - E(t) + R(t) + U(t) \tag{12}$$

$$J(h, E, U) = \alpha \int (h - h_0)^2 \,\mathrm{d}t - \beta \int E(t)\,\mathrm{d}t - \gamma \int U(t)\,\mathrm{d}t \tag{13}$$

where

- t is the time;
- h_i is the water level of lake i;
- I is the inflow from others;
- O is the outflow to others;
- E is the water evaporation;
- R is the rainfall;
- U is the human control.